

Semi-analytical estimates for overall response of porous elastomer filled with power-law fluid

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ABSTRACT

Semi-analytical estimates are derived for effective behavior of porous elastomeric composites, containing square arrangement of mono-disperse circular pores, filled with a non-Newtonian, power-law fluid. The elastomeric composite is kept in a reservoir of same fluid, such that the fluid from pores drains into a reservoir as the elastomer is compressed and upon relaxation of load the liquid gets sucked back into the pores. The estimates are compared with corresponding results obtained from numerical simulations. Finite element analysis software COMSOL multiphysics was used for the numerical simulations. The estimates for stored energy density and dissipation density of the composite match well with the simulation results. Stress-strain curves for the power-law fluid-filled composites are plotted under monotonic and harmonic loadings. Equivalent spring stiffness and damping coefficient, associated with uniaxial plane strain loading of these composites, are determined as a function of material properties of the elastomer and power-law fluid, and microstructural parameters of the unit cell. The derived expressions can be used to develop custom composite materials for suspension, impact protection, and vibration isolation applications. The present work is an extension of a previous work by the authors on overall response of Newtonian fluid-filled elastomers.

1. Introduction

Energy absorption and vibration damping are important in automobiles [1,2], bullet proof vests [3], sports equipment [4], protection equipment [5], etc. Elastomeric foams have widely been used as soft materials for energy absorption [6,7] as they can undergo recoverable deformations. However, when subjected to large deformations the cell-walls can get damaged and complete recovery of deformation may not be possible. Therefore, foams need to be supplemented to enhance their strength and improve their energy absorption capacity. Fluid-filled elastomers that drain/absorb the fluid as the elastomers are deformed are excellent candidates for such applications. In the present study, closed-form expressions are derived for power-law fluid filled elastomer in terms of initial pore radius, initial porosity, fluid consistency index, fluid power-law index, elastomer parameters and loading rate.

Elastomeric foam (without any fluid) are used in packaging, noise reduction, automobiles, furniture and energy absorption. Macroscopic response of porous elastomers depends on loading type, material property and pore microstructure. Multiple phenomenological [8] and homogenization based [9–13] constitutive models have been proposed to model the macroscopic response of porous elastomers. Models based on spherical and cylindrical assemblages [12,13] have also been proposed to describe their behavior. Several models have been proposed to predict material behavior of porous viscoelastic materials [14–16].

While there are a number of models for estimating the effective macroscopic response of porous elastomers and porous viscoelastic materials, there is no micro-mechanics based or homogenization based study to estimate the macroscopic response of power-law fluid-filled elastomers taking into account the porosity, elastomer properties, power-law fluid behavior and nature of loading. Nonetheless, there is an earlier work from the authors of this paper on behavior of elastomers filled with Newtonian fluids [17].

Extensive experimental studies have been carried out for shear-thickening fluid (STF) impregnated fabrics for enhanced impact protection as compared to traditional materials. Shear-thickening fluids (power-law fluids with power-law index $n > 1$) can undergo multiple orders of viscosity increase at critical shear rate and given stress [18]. Silica dispersed in ethylene glycol is one of the most commonly used shear-thickening fluid. Addition of shear-thickening fluid to bullet-proof vests [19] enhanced their ballistic protection while it also provided flexibility. Experiments [20,21] have been used to study the response of STF filled elastomers. Response of the composite is strongly affected by the strain rate. At low strain rates it has a fluid like response whereas at high strain rates the fluid stiffens and its energy absorption improves. STF filled Kevlar fibers and warp-knitted spacer fibers (WKSF) have been investigated for stab resistance [22] and low velocity impact protection [23], respectively. Performance of both composites

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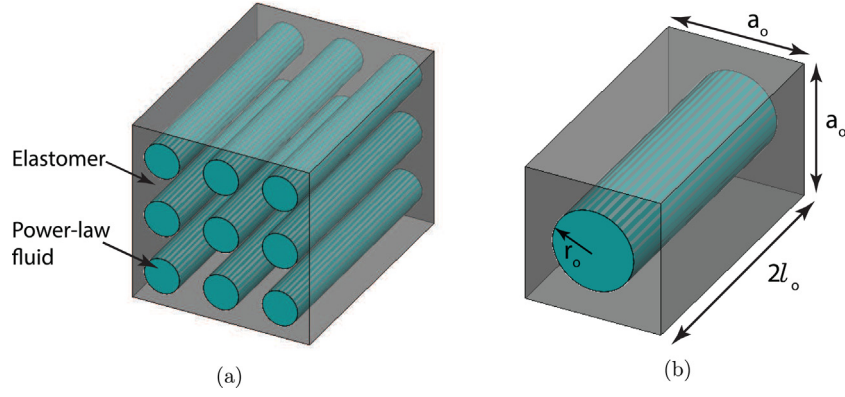


Fig. 1. Schematic of porous elastomer with cylindrical pores arranged in square array. (a) Square arrangement of mono-dispersed cylindrical pores in the elastomer. The cylindrical pores are completely filled with a power-law fluid. (b) Unit cell used for analytical study and numerical simulation [17].

improved with the addition of STF. Soutrenon and Michaud [18] experimentally studied the effectiveness of a shear-thickening fluid filled open-cell melamine foam in impact protection. Energy absorption of up to 85% and considerable reduction in the force transmitted was observed. Haris et al. [24] showed that addition of STF enhances shock wave mitigation performance. Impregnation of shear-thickening fluid reduced the normalized average peak pressure transmitted from 2.46 to 1.49 while the normalized maximum rate of pressure rise got reduced significantly from 2.3 to 0.76. Xiaoke et al. [25] experimentally showed that shear-thickening gel impregnated polyurethane foam provided enhanced impact protection. Turtoi et al. [26] experimentally and analytically studied the flow of Bingham fluid, when the fluid filled reticulated foam is subjected to compression. Addition of Bingham fluid to the foam caused stiffening of the composite. Sen et al. [27] carried out numerical simulations to study the effect of arrangement of Kevlar fabric and shear-thickening fluid impregnated Kevlar fabric were considered. Effect of projectile shape, impact velocity and shear-thickening fluid characteristics were investigated.

In this work, a micromechanics based model is formulated to get a better understanding of the response of a porous elastomer filled with a power-law fluid. Three dimensional model of an unit cell consisting of a porous elastomer completely filled with a power-law fluid is constructed. The elastomer contains square arrangement of cylindrical pores. Closed-form expressions are derived for the following: (i) viscous dissipation due to deformation induced flow in the power-law fluid, (ii) macroscopic stress-strain behavior of the power-law fluid filled elastomer under monotonic and harmonic loadings. For a special loading case of uniaxial plane strain loading, the composite is assumed to behave like a Kelvin-Voigt material, and hence expression for its equivalent spring stiffness and damping coefficient are derived. The accuracy of estimates obtained using closed-form expressions is validated by comparing the estimates with corresponding results obtained using numerical simulation. Numerical simulations have been carried out using the commercial finite element analysis software COMSOL multiphysics. To solve for strain energy density and average deformation gradient in the porous elastomer, we use the concept of cylindrical assemblage and solve an equivalent problem for the assemblage with identical porosity. The velocity field for the fluid is estimated using the weighted integral approach.

2. Preliminaries

2.1. Microstructure details

Microstructure of the porous elastomer considered in this work is shown in Fig. 1(a). Square arrangement of the mono-dispersed cylindrical pores is studied. Fig. 1(b) shows the unit cell geometry used for

analytical studies and numerical simulation in COMSOL. Pore initial radius r_0 , initial porosity f_0 and initial pore length $2l_0$ characterize the geometry of the unit cell.

Analytical estimates have been derived for power-law fluid filled elastomer subjected to plane strain loadings. A special case of this is uniaxial plane strain loading (see Fig. 2), where the behavior of power-law fluid filled elastomer is similar to that of a spring-damper system. The bracket inside which the elastomer is kept has vertical slits to allow free movement of fluid in and out of the pore and at the same time it restricts the movement of elastomer in every direction other than the loading direction. Therefore, the presence of bracket causes the elastomer to undergo uniaxial plane strain compression. The elastomer and bracket assembly is placed inside a container filled with the same power-law fluid as that saturating the pores. During compression, the pore size reduces which causes the fluid to flow out into the outer container. When the load is released, pores regain their original size and fluid flows back into the pores.

2.2. Constituent properties

2.2.1. Fluid properties

The pores are fully filled with a non-Newtonian, power-law, incompressible fluid. The apparent dynamic viscosity $\eta(\dot{\gamma})$ for the power-law fluid is taken as

$$\eta(\dot{\gamma}) = m|\dot{\gamma}|^{(n-1)}, \quad n > 0 \quad (1)$$

where m is the consistency factor, n is the power-law index and $|\dot{\gamma}| = \sqrt{\frac{1}{2}\dot{\gamma} : \dot{\gamma}}$ where $\dot{\gamma} = (\nabla \mathbf{v} + (\nabla \mathbf{v})^T)$ with $\mathbf{v}$ being fluid velocity. Navier-Stokes equation and the continuity equation govern the flow of fluid. Here, ρ_f is fluid density and p is fluid pressure. The primary variables to be determined in the fluid phase are pressure p and fluid velocity $\mathbf{v}$.

2.2.2. Elastomer properties

In the present study, the elastomer is taken as an incompressible, Neo-Hookean hyperelastic material with strain energy density function:

$$W^{(s)}(\mathbf{F}) = \frac{G}{2} (I_1 - 3) \quad (2)$$

where G is the shear modulus of elastomer in initial state, $I_1 = \text{tr}(\mathbf{C})$ is the first principal invariant of right Cauchy-Green deformation tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ and $\mathbf{F}$ is the deformation gradient tensor. Closed form analytical expressions can be derived for macroscopic strain energy density for porous Neo-Hookean hyperelastic elastomers. Other I_1 based strain energy density functions such as Gent [28,29], Yeoh [28] or I_1, I_2 based strain-energy density functions such as Gent-Gent [30] can be used for analysis of porous elastomeric composites, particularly for large deformations. However, closed form analytical solutions for macroscopic strain energy density of porous elastomers would not be

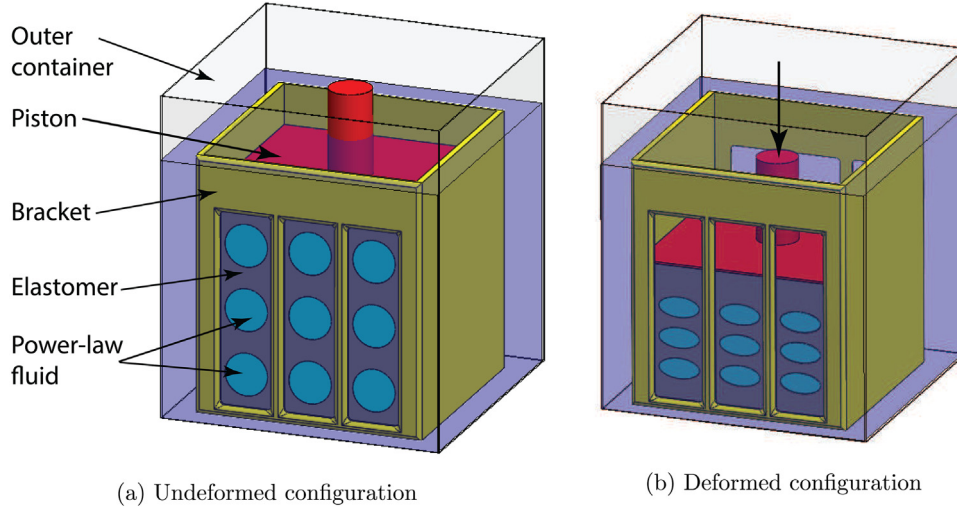


Fig. 2. Schematic of the assembly showing power-law fluid filled elastomer subjected to uniaxial plane-strain loading. (a) Undeformed/initial state of elastomer and (b) deformed/final configuration of the elastomer [17].

feasible for these models. Numerical results would have to be obtained. Focus in this work, has been primarily for small deformations, where Neo-Hookean model is applicable. Inertial effects and body forces have been neglected in the present study, the equilibrium equation for the solid is taken as

$$\nabla \cdot \sigma = 0 \quad (3)$$

is the Cauchy stress tensor. Displacement is the primary variable to be determined for solid phase.

2.3. Boundary conditions

2.3.1. Fluid phase

In the present study, upon application of load the elastomer deforms which causes the fluid to flow in/out of the pore, therefore this is a case of deformation induced flow. No-slip condition is applied at the solid–fluid interface to impose velocity continuity and pore ends are kept at constant pressure.

2.3.2. Solid phase

Affine displacement boundary condition i.e. displacement of the form

$$\mathbf{u} = (\bar{\mathbf{F}} - \mathbf{I}) \mathbf{X}, \quad (4)$$

is applied at the outer boundary of the representative volume element (RVE). Here $\bar{\mathbf{F}}$ is the applied macroscopic deformation gradient. In the present study, closed-form expressions are obtained for plane strain deformations of the form

$$\bar{\mathbf{F}} = \begin{bmatrix} \bar{\lambda}_1 & 0 & 0 \\ 0 & \bar{\lambda}_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (5)$$

The porous elastomer due to the presence of pores is macroscopically compressible, even though the elastomer is microscopically incompressible, therefore $\det \bar{\mathbf{F}} \neq 1$. We would like to specify that terms with only bar represent average macroscopic property, terms with ‘s’ in superscript represent local property for solid phase and similarly terms having superscript ‘f’ denote local property for fluid phase.

2.4. Numerical modeling

The estimates obtained using closed-form expressions are validated using a commercial finite element analysis software COMSOL multi-physics for carrying out coupled fluid–structure interaction simulations.

Table 1

Mesh parameters of the mesh used for numerical simulation in COMSOL.

Parameters	Mesh
No. of domain elements	184883
No. of boundary elements	22738
No. of edge elements	972
Min. solid element size (mm)	0.18
Max. solid element size (mm)	1.00
Min. fluid element size (mm)	0.0436
Max. fluid element size (mm)	0.135
Boundary layers	2

Full Navier–Stokes equation including the inertial terms is considered for fluid flow. Fluid domain has moving boundaries (pore wall), therefore it is considered as a deforming domain. Grid independence was analyzed by considering different mesh sizes for initial porosity $f_o = 0.6$ and initial pore radius $r_o = 0.5$ mm. Properties of the mesh considered in numerical simulations are given in Table 1.

Domain discretization and solver related details are given below

1. Domain discretization

Solid domain: Quadratic serendipity elements

Fluid domain: Second order elements for velocity and linear element for pressure

2. Time stepping scheme: Backward differentiation formula (BDF)

3. Solver used: PARDISO

4. Relative tolerance for convergence = 0.005

5. Formulation

Solid domain: Lagrangian

Fluid domain: Eulerian

Fluid–solid interface: Arbitrary Lagrangian–Eulerian

Mesh convergence was carried out for three different mesh sizes. The given mesh was selected because the results did not vary with further reduction in mesh size.

3. Estimates for effective properties of the composite

3.1. Constitutive model for porous elastomer

Composite cylindrical assemblage model was first used by Hashin and Rosen [31] to estimate the effective stiffness of aligned hollow fiber reinforced composites. Composite cylindrical assemblage cannot capture the exact microstructure of the composite, but it has been shown using numerical simulations that cylindrical assemblage provides very

good estimate for strain energy [32]. A thick-walled annular cylinder, in cylindrical assemblage, is considered in the initial undeformed porous elastomer. The volume fraction of pore in the assemblage is identical to porosity in the composite. This cylindrical assemblage is subjected to deformation gradient $\bar{\mathbf{F}}$, which consists of principal macroscopic stretches $\bar{\lambda}_1, \bar{\lambda}_2, \bar{\lambda}_3 = 1$. For a more general case $\bar{\lambda}_1 \neq \bar{\lambda}_2 \neq \bar{\lambda}_3$.

A kinematically admissible deformation field similar to that proposed in [33], is taken such that (origin is considered at center of annular cylinder)

$$\mathbf{x} = \psi_i(R)\mathbf{X}, \quad R = \sqrt{X_1^2 + X_2^2} \quad (i = 1, 2) \text{ and no sum on } i \quad (6)$$

where $\mathbf{x}$ and $\mathbf{X}$ denote the initial and final location of a point respectively and R denotes the distance of the point from center of the pore in initial state. Therefore, the microscopic deformation gradient in hollow cylinder is written as

$$\mathbf{F}_{ij}^{(s)} = \psi_i' \frac{X_i X_j}{R} + \psi_i \delta_{ij}, \quad i, j = 1, 2 \quad (\text{no sum on } i) \quad (7)$$

where $O' = \frac{\partial(\cdot)}{\partial R}$. The elastomer considered in the present study is microscopically incompressible, therefore we have

$$\det \mathbf{F}^{(s)} = 1. \quad (8)$$

Combining (7) and (8) results in a set of equations, solved by using the boundary condition that the outer boundary of hollow cylinder is subjected to $\bar{\mathbf{F}}$. Therefore, the microscopic deformation gradient tensor $\mathbf{F}^{(s)}$ in the hollow cylinder, is written as

$$\mathbf{F}^{(s)} = \begin{bmatrix} \frac{\bar{\lambda}_1}{\bar{J}^{1/2}} \left(\frac{1-\psi^2}{\psi R^2} X_1^2 + \psi \right) & \frac{\bar{\lambda}_1}{\bar{J}^{1/2}} \left(\frac{1-\psi^2}{\psi R^2} X_1 X_2 \right) & 0 \\ \frac{\bar{\lambda}_2}{\bar{J}^{1/2}} \left(\frac{1-\psi^2}{\psi R^2} X_1 X_2 \right) & \frac{\bar{\lambda}_2}{\bar{J}^{1/2}} \left(\frac{1-\psi^2}{\psi R^2} X_2^2 + \psi \right) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (9)$$

where $\psi = \left[1 + (R_0/R)^2 (\bar{J} - 1) \right]^{1/2}$. Using (9), $I_1^{(s)}$ is obtained as

$$I_1^{(s)} = \frac{1}{\bar{J}} \left\{ (\bar{I}_1 - 1)\psi^2 + \frac{1}{R^2} \left(\frac{1}{\psi^2} - \psi^2 \right) (\bar{\lambda}_1^2 X_1^2 + \bar{\lambda}_2^2 X_2^2) \right\} + 1. \quad (10)$$

Therefore, the stored energy density in the composite $\bar{W}_{SE}$ is obtained as

$$\bar{W}_{SE} = \frac{1}{V_0} \int_{V_s} W^{(s)} dV. \quad (11)$$

where V_0 is the initial volume of composite in undeformed state and V_s is the current volume of elastomer in the deformed state. Using (2) and (11), the stored energy density in the composite is estimated as [17]

$$\bar{W}_{SE}(\bar{\mathbf{F}}) = \frac{G}{2} \frac{(\bar{I}_1 - 1)(\bar{J} - 1)}{2\bar{J}} \log \left(\frac{f_0 + \bar{J} - 1}{f_0 \bar{J}} \right) + G(1 - f_0) \left(\frac{(\bar{I}_1 - 1)}{2\bar{J}} - 1 \right) \quad (12)$$

where $\bar{I}_1 = \text{tr}(\bar{\mathbf{C}})$, right Cauchy–Green deformation tensor $\bar{\mathbf{C}} = \bar{\mathbf{F}}^T \bar{\mathbf{F}}$, where $\bar{\mathbf{F}}$ is the macroscopic deformation gradient tensor, $\bar{J} = \det \bar{\mathbf{F}}$ and f_0 is initial porosity in the elastomer.

3.2. Dissipation due to deformation induced flow

Velocity at the solid–fluid interface is evaluated as

$$\mathbf{v} = \frac{\partial \mathbf{u}}{\partial t} = \bar{\mathbf{L}}^{(f)} \mathbf{x} \quad (13)$$

where

$$\bar{\mathbf{L}}^{(f)} = \dot{\bar{\mathbf{F}}} \bar{\mathbf{F}}^{(f)-1} = \begin{bmatrix} \dot{\bar{\lambda}}_1 / \bar{\lambda}_1 & 0 & 0 \\ 0 & \dot{\bar{\lambda}}_2 / \bar{\lambda}_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (14)$$

with

$$\dot{\bar{\lambda}}_1^{(f)} = \bar{\lambda}_1 \sqrt{\frac{(f_0 + \bar{J} - 1)}{f_0 \bar{J}}}, \quad \dot{\bar{\lambda}}_1^{(f)} = \frac{2\dot{\bar{\lambda}}_1 \bar{J}(f_0 + \bar{J} - 1) + \bar{\lambda}_1(1 - f_0)\dot{\bar{J}}}{2f_0^{1/2} \bar{J}^{3/2} \sqrt{f_0 + \bar{J} - 1}}, \quad (15)$$

$$\dot{\bar{\lambda}}_2^{(f)} = \bar{\lambda}_2 \sqrt{\frac{(f_0 + \bar{J} - 1)}{f_0 \bar{J}}}, \quad \dot{\bar{\lambda}}_2^{(f)} = \frac{2\dot{\bar{\lambda}}_2 \bar{J}(f_0 + \bar{J} - 1) + \bar{\lambda}_2(1 - f_0)\dot{\bar{J}}}{2f_0^{1/2} \bar{J}^{3/2} \sqrt{f_0 + \bar{J} - 1}}. \quad (16)$$

Here $\dot{\bar{\lambda}}_1^{(f)}, \dot{\bar{\lambda}}_2^{(f)}, \dot{\bar{\lambda}}_3^{(f)} = 0$ are the average rate of stretch for the pore along principal directions. Using Eqs. (13), (14) and considering no-slip at solid–fluid interface, kinematically admissible velocity field for the fluid phase is proposed as [17]

$$v_1 = \frac{\dot{\bar{\lambda}}_1^{(f)} x_1}{\bar{\lambda}_1^{(f)}} \left(A \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_0^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_0^2} \right) + 1 \right) \quad (17a)$$

$$v_2 = \frac{\dot{\bar{\lambda}}_2^{(f)} x_2}{\bar{\lambda}_2^{(f)}} \left(B \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_0^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_0^2} \right) + 1 \right) \quad (17b)$$

$$v_3 = C \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_0^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_0^2} \right) \frac{x_3}{l_0} \quad (17c)$$

where A, B, C are constants and $\bar{\lambda}_1^{(f)}, \bar{\lambda}_2^{(f)}$ are components of the average deformation gradient tensor for the pore (fluid) phase $\bar{\mathbf{F}}^{(f)} = \text{diag}(\bar{\lambda}_1^{(f)}, \bar{\lambda}_2^{(f)}, 1)$. $(\cdot)' = \frac{\partial(\cdot)}{\partial t}$. We solve for C using the condition that at any distance x_3 from the center of pore the volume of fluid leaving the cross-section is equal to the change in volume of pore for that length. Therefore, we get

$$C = -2l_0 \left(\frac{\dot{\bar{\lambda}}_1^{(f)}}{\bar{\lambda}_1^{(f)}} + \frac{\dot{\bar{\lambda}}_2^{(f)}}{\bar{\lambda}_2^{(f)}} \right). \quad (18)$$

Thereafter, least-square method is used with the continuity equation to get A and B as

$$A = -\frac{C \bar{\lambda}_1^{(f)}}{4l_0 \bar{\lambda}_1}, \quad \text{and} \quad B = -\frac{C \bar{\lambda}_2^{(f)}}{4l_0 \bar{\lambda}_2}. \quad (19)$$

Cauchy stress tensor at any point in the fluid is given as

$$\boldsymbol{\sigma} = -p \mathbf{I} + \eta(\dot{\gamma}) \dot{\gamma} \quad (20)$$

where p is fluid pressure, $\mathbf{I}$ is identity tensor, $\eta(\dot{\gamma})$ is the apparent viscosity of the power-law fluid. Using (1) and (20) the local dissipation density is estimated as

$$\dot{W}^{(f)}(\dot{\gamma}) = \frac{1}{2} \boldsymbol{\sigma} : \dot{\gamma} = \frac{1}{2} \eta(\dot{\gamma}) \dot{\gamma} : \dot{\gamma} = m \left(\frac{\dot{\gamma} : \dot{\gamma}}{2} \right)^{\frac{(n+1)}{2}} \quad (21)$$

where

$$\dot{\gamma} : \dot{\gamma} = 4 \left(\frac{\partial v_1}{\partial x_1} \right)^2 + 4 \left(\frac{\partial v_2}{\partial x_2} \right)^2 + 4 \left(\frac{\partial v_3}{\partial x_3} \right)^2 + 2 \left[\left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_1} \right)^2 + \left(\frac{\partial v_1}{\partial x_3} + \frac{\partial v_3}{\partial x_1} \right)^2 + \left(\frac{\partial v_2}{\partial x_3} + \frac{\partial v_3}{\partial x_2} \right)^2 \right].$$

Therefore, the expression for local viscous dissipation in the power-law fluid becomes

$$\dot{W}^{(f)} = m 2^{\frac{(n+1)}{2}} \left(\left(\frac{\partial v_1}{\partial x_1} \right)^2 + \left(\frac{\partial v_2}{\partial x_2} \right)^2 + \left(\frac{\partial v_3}{\partial x_3} \right)^2 + \frac{1}{2} \left[\left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_1} \right)^2 + \left(\frac{\partial v_1}{\partial x_3} + \frac{\partial v_3}{\partial x_1} \right)^2 + \left(\frac{\partial v_2}{\partial x_3} + \frac{\partial v_3}{\partial x_2} \right)^2 \right] \right)^{\frac{(n+1)}{2}}. \quad (22)$$

Using (22) the dissipation density in the composite, $\dot{\bar{W}}_{VD}$ is calculated as

$$\dot{\bar{W}}_{VD} = \frac{1}{V_0} \int_{V_f} \dot{W}^{(f)} dV \quad (23)$$

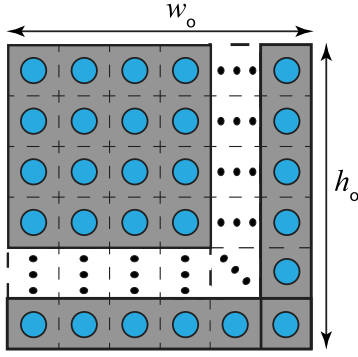


Fig. 3. Schematic representation of fluid-filled elastomer having square arrangement of $M \times N$ unit-cells.

where V_f is current volume of the fluid domain and V_0 is the initial volume of composite. Since the expression for local viscous dissipation density is complicated, the analytical expression for dissipation density of the composite is also complicated.

4. Uniaxial plane strain loading

In the previous sections the expressions for $\bar{\mathbf{F}}^{(s)}$, $\bar{W}_{SE}$, $\bar{\mathbf{F}}^{(f)}$ and $\dot{\bar{W}}_{VD}$ were derived for a more general case of plane strain loading. Now, in this section results are obtained specifically for the case where the porous elastomer having mono-dispersed cylindrical pores, fully saturated with power-law fluid, is subjected to uniaxial plane strain loading (see Fig. 2). Uniaxial plane strain loading is characterized by applied deformation occurring only in the vertical direction such that $\bar{\lambda}_1 = 1$ and $\bar{\lambda}_2 = \bar{\lambda}$. Therefore, the applied macroscopic deformation gradient tensor $\bar{\mathbf{F}}$ becomes

$$\bar{\mathbf{F}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \bar{\lambda} & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (24)$$

4.1. Stored energy density of the composite

Composite's stored energy density $\bar{W}_{SE}$ (energy stored in porous elastomer filled with power-law fluid per unit initial volume of the composite) for the case of uniaxial plane strain loading is obtained by substituting the principal macroscopic stretch values from (24) in (12). Therefore the expression for $\bar{W}_{SE}$ becomes

$$\begin{aligned} \bar{W}_{SE}(\bar{\mathbf{F}}) = & \frac{G}{2} \frac{(\bar{\lambda}^2 + 1)(\bar{\lambda} - 1)}{2\bar{\lambda}} \log \left(\frac{f_0 + \bar{\lambda} - 1}{f_0 \bar{\lambda}} \right) \\ & + G(1 - f_0) \left(\frac{\bar{\lambda}^2 - 2\bar{\lambda} + 1}{2\bar{\lambda}} \right). \end{aligned} \quad (25)$$

Taking time derivative of (25), we get $\dot{\bar{W}}_{SE}$ as

$$\begin{aligned} \dot{\bar{W}}_{SE}(\bar{\mathbf{F}}) = & \frac{G\dot{\bar{\lambda}}}{2\bar{\lambda}^2} \left[(1 - f_0) \left((\bar{\lambda}^2 - 1) + \frac{(\bar{\lambda}^2 + 1)(\bar{\lambda} - 1)}{2(f_0 + \bar{\lambda} - 1)} \right) \right. \\ & \left. + \log \left(\frac{f_0 + \bar{\lambda} - 1}{f_0 \bar{\lambda}} \right) \frac{2\bar{\lambda}^3 - \bar{\lambda}^2 + 1}{2} \right]. \end{aligned} \quad (26)$$

4.2. Dissipation density of the composite

The principal macroscopic stretches are substituted in (17a), (17b), (17c) to get the expressions for velocity components, which are then

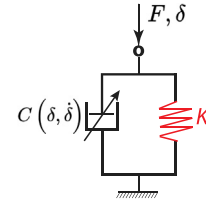


Fig. 4. Generalized Kelvin-Voigt model for composite.

substituted in (22) to get the local viscous dissipation density under uniaxial plane strain loading as

$$\begin{aligned} \dot{W}^{(f)} = & m 2^{\frac{(n+1)}{2}} \left(\frac{|\dot{\bar{\lambda}}|}{f_0 + \bar{\lambda} - 1} \right)^{(n+1)} \\ & \times \left(\frac{(1 - f_0)^2}{4\bar{\lambda}^2} \left[1 + A \left(1 - \frac{3x_1^2}{\bar{\lambda}_1^{(f)2} r_0^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_0^2} \right) \right]^2 \right. \\ & + \frac{(f_0 + 2\bar{\lambda} - 1)^2}{4\bar{\lambda}^2} \left[1 + B \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_0^2} - \frac{3x_2^2}{\bar{\lambda}_2^{(f)2} r_0^2} \right) \right]^2 \\ & + 4\bar{\lambda} \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_0^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_0^2} \right)^2 \\ & \left. + \frac{f_0^2}{2r_0^4(f_0 + \bar{\lambda} - 1)^2} \left[x_1^2 x_2^2 \left(\bar{\lambda} + \frac{1}{\bar{\lambda}} \right)^2 + 16x_3^2 \left(x_1^2 \bar{\lambda}^2 + \frac{x_2^2}{\bar{\lambda}^2} \right) \right] \right)^{\frac{(n+1)}{2}}. \end{aligned} \quad (27)$$

Substituting (27) in (22), the dissipation density of the composite is obtained as

$$\dot{\bar{W}}_{VD} = m 2^{\frac{(n+1)}{2}} \left(\frac{|\dot{\bar{\lambda}}|}{f_0 + \bar{\lambda} - 1} \right)^{(n+1)} \frac{I}{V_0} \quad (28)$$

where I is defined as

$$\begin{aligned} I = & \int_{V_f} \left(\frac{(1 - f_0)^2}{4\bar{\lambda}^2} \left[1 + A \left(1 - \frac{3x_1^2}{\bar{\lambda}_1^{(f)2} r_0^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_0^2} \right) \right]^2 \right. \\ & + \frac{(f_0 + 2\bar{\lambda} - 1)^2}{4\bar{\lambda}^2} \left[1 + B \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_0^2} - \frac{3x_2^2}{\bar{\lambda}_2^{(f)2} r_0^2} \right) \right]^2 \\ & + 4\bar{\lambda} \left(1 - \frac{x_1^2}{\bar{\lambda}_1^{(f)2} r_0^2} - \frac{x_2^2}{\bar{\lambda}_2^{(f)2} r_0^2} \right)^2 \\ & \left. + \frac{f_0^2}{2r_0^4(f_0 + \bar{\lambda} - 1)^2} \left[x_1^2 x_2^2 \left(\bar{\lambda} + \frac{1}{\bar{\lambda}} \right)^2 + 16x_3^2 \left(x_1^2 \bar{\lambda}^2 + \frac{x_2^2}{\bar{\lambda}^2} \right) \right] \right)^{\frac{(n+1)}{2}} dV. \end{aligned} \quad (29)$$

The integral I is estimated numerically, by discretizing the fluid volume using 1st order hexahedral elements, and estimating the integral over each element using Gauss-Legendre quadrature scheme with $2 \times 2 \times 2$ integration points.

4.3. Small strain estimates

Considering the deformation to be small ($\bar{\epsilon} = \bar{\lambda} - 1 \rightarrow 0$), stored energy density of the composite and its time derivative becomes

$$\bar{W}_{SE}(\bar{\mathbf{F}}) = \frac{G}{2} \frac{(1 - f_0^2)}{f_0} \bar{\epsilon}^2 \quad \text{and} \quad \dot{\bar{W}}_{SE}(\bar{\mathbf{F}}) = G \frac{(1 - f_0^2)}{f_0} \bar{\epsilon} \dot{\bar{\epsilon}}. \quad (30)$$

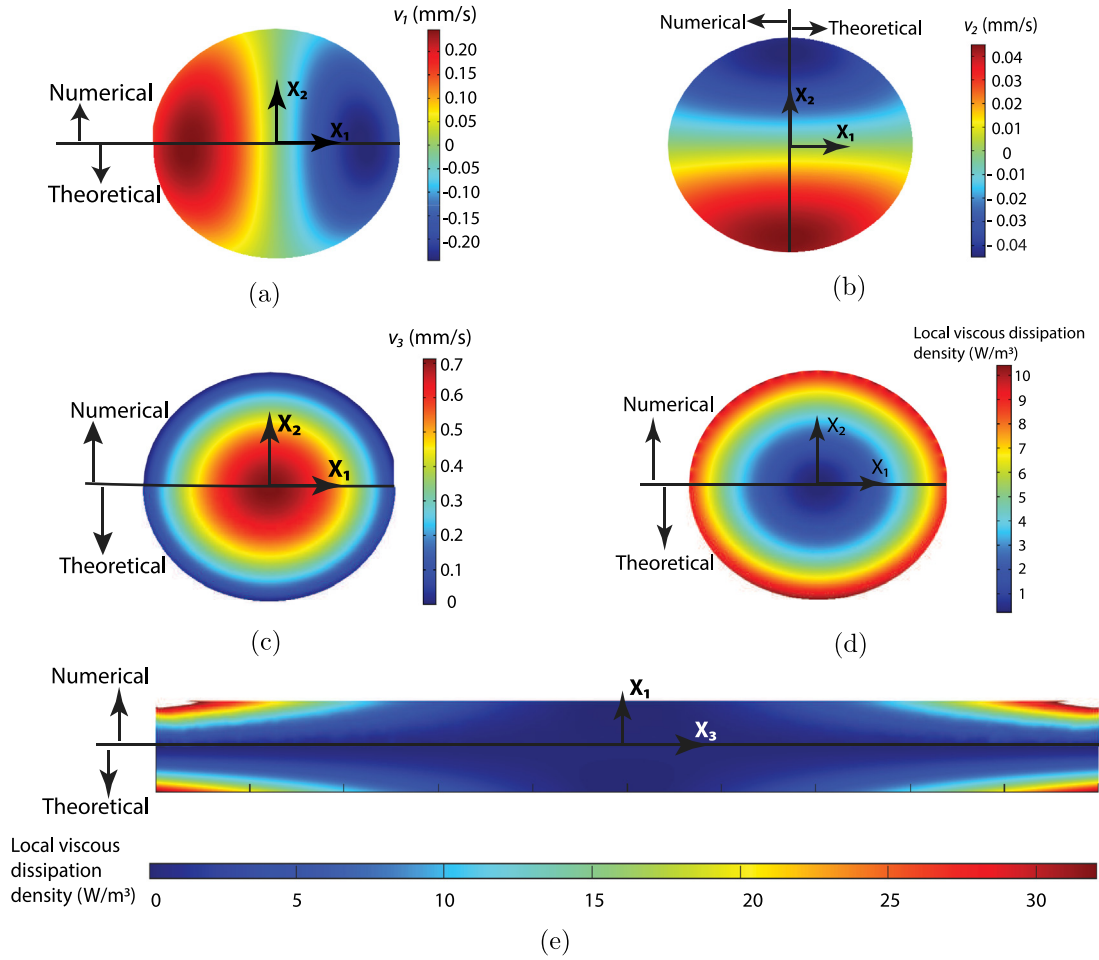


Fig. 5. Analytical estimate and numerical simulation result comparison for velocity components (a) v_1 , (b) v_2 , (c) v_3 and (d, e) local viscous dissipation density in x_1x_2 and x_1x_3 plane respectively. x_1x_2 plane is located at $x_3 = l_o/2$ and x_1x_3 plane is located at $x_2 = 0$. The results are plotted for $\bar{\lambda} = 0.97$, $n = 1.3$, $f_o = 40\%$, $r_o = 0.5$ mm, $a_o = 1.4012$ mm and $\dot{\bar{\lambda}} = -0.05/s$.

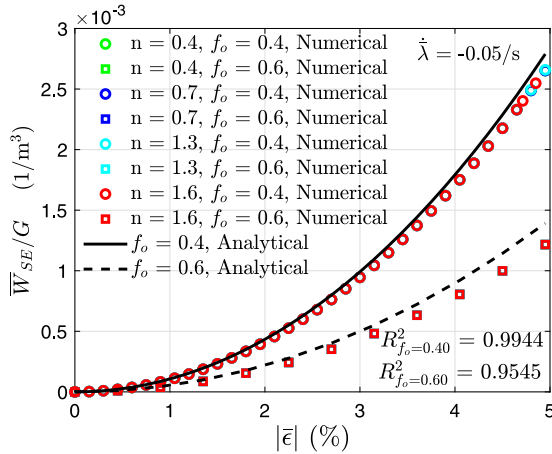


Fig. 6. Normalized stored energy density of power-law fluid filled elastomeric composite subject to uniaxial plane-strain compression as obtained from numerical simulation and analytical estimates. Results are plotted for initial void fraction $f_o = 0.4$ and 0.6 . Solid, dashed lines denote analytical results and square, circular markers denote numerical results.

For the case when the deformation is small, the dissipation density in the composite becomes

$$\bar{W}_{VD} = m 2^{\frac{(n+1)}{2}} \frac{|\dot{\bar{\epsilon}}|^{(n+1)} I}{f_o^{(n+1)} V_0} = m 2^{\frac{(n+1)}{2}} \frac{|\dot{\bar{\epsilon}}|^{(n+1)} I}{2 f_o^n \pi r_o^2 l_o}. \quad (31)$$

4.4. Macroscopic behavior of the power-law fluid filled elastomer

Equation for power balance per unit initial volume of the composite is written as

$$\dot{\bar{W}} = \dot{\bar{W}}_{SE} + \dot{\bar{W}}_{VD} \quad (32)$$

where $\dot{\bar{W}}$ is the external power input. Assuming strains to be small and plane strain deformation, we have

$$\dot{\bar{W}} = \bar{\sigma} \dot{\bar{\epsilon}}. \quad (33)$$

Therefore, we get

$$\bar{\sigma} = G \frac{(1 - f_o^2)}{f_o} \bar{\epsilon} + \alpha |\dot{\bar{\epsilon}}|^{(n-1)} \dot{\bar{\epsilon}} \quad (34)$$

where

$$\alpha = \frac{m 2^{\frac{(n-1)}{2}} I}{f_o^n \pi r_o^2 l_o}.$$

For a given macroscopic displacement δ , macroscopic strain and strain rate is given as

$$\bar{\epsilon} = \frac{\delta}{h_o} \quad \text{and} \quad \dot{\bar{\epsilon}} = \frac{\dot{\delta}}{h_o}, \quad (35)$$

and cross-sectional area for the foam (see Fig. 3) for applied load is $2Ma_o l_o$. Re-writing the (34) in terms of force and displacement, considering generalized Kelvin-Voigt model (see Fig. 4) we have

$$F = \frac{2G w_o l_o (1 - f_o^2)}{h_o f_o} \delta + \frac{2\alpha w_o l_o |\dot{\delta}|^{(n-1)}}{h_o^n} \dot{\delta} \quad (36)$$

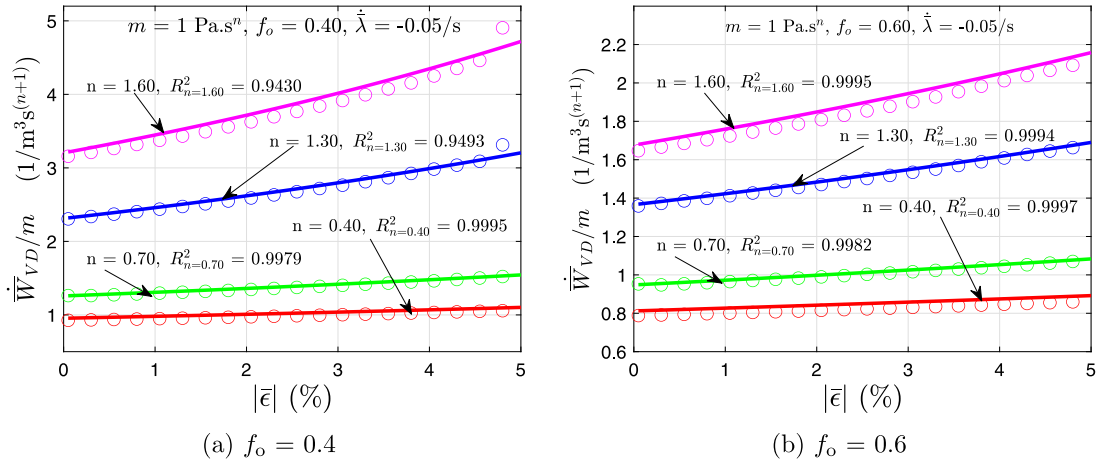


Fig. 7. Theoretical estimates and numerical simulation results for dissipation density in the composite under uniaxial plane strain compression. The shown results are for a power-law fluid having consistency index $m = 1 \text{ Pa s}^n$, power-law index values of $n = 0.4, 0.7, 1.3$ and 1.6 and loading rate $\dot{\lambda} = -0.05/\text{s}$. Solid lines denote theoretical estimates and circular markers denote numerical results.

where δ is the applied displacement, $\dot{\delta}$ the velocity and F is the resultant force.

The governing equation for a single degree of freedom spring-damper system is of the form

$$F = K \delta + C \dot{\delta} \quad (37)$$

where K is the stiffness of the spring and C is the damping coefficient. Comparing (36) and (37), we get

$$K = \frac{2 G w_o l_o (1 - f_o^2)}{h_o f_o}, \quad C = \frac{2 \alpha w_o l_o}{h_o^n} |\dot{\delta}|^{(n-1)}. \quad (38)$$

Thus, the damping coefficient in power-law fluid filled elastomer depends on loading rate, in addition to material and geometric parameters.

4.4.1. Harmonic loading

When subjected to a harmonic input of the form $\delta = \delta_o \sin \omega t$, the equivalent linear stiffness K_h and equivalent damping coefficient C_h are defined as the coefficients of the fundamental harmonics of the restoring force.

$$K_h = \frac{\omega}{\delta_o \pi} \int_0^\tau F \sin \omega t dt \quad \text{and} \quad C_h = \frac{1}{\delta_o \pi} \int_0^\tau F \cos \omega t dt \quad (39)$$

where τ is the time period of harmonic input. For power-law fluid filled elastomers, subjected to harmonic loadings under uniaxial plane strain loading, we get

$$K_h = \frac{2 G w_o l_o (1 - f_o^2)}{h_o f_o}, \quad C_h = \frac{8 \alpha w_o l_o \delta_o^{(n-1)} \omega^n}{\pi h_o^n} \int_0^{\pi/2\omega} (\cos \omega t)^{(n+1)} dt. \quad (40)$$

5. Results and discussion

Elastomer and fluid properties chosen in numerical and analytical analysis, correspond to that of polyurethane [34,35] and glycerine [36], respectively.

5.1. Validation of velocity and viscous dissipation density function

Fig. 5 shows contour plots for velocity components (v_1, v_2 and v_3) and local viscous dissipation density in the power-law fluid when the composite is subjected to monotonic compression loading. Analytical estimates and numerical simulation results have been plotted alongside for comparison. Fig. 5(a), 5(b) and 5(c) show the comparison of contour plots for v_1, v_2, v_3 in the $x_1 x_2$ plane at $x_3 = l_o/2$, $\bar{\lambda} = 0.97$. It can be

seen from the plots that the velocity field is very well captured by the proposed velocity functions. It can be seen from Figs. 5(d) and 5(e) that local viscous dissipation density function proposed in (27) accurately captures the viscous dissipation in power-law fluid.

5.2. Validation of stored energy density of composite

Fig. 6 shows the evolution of stored energy density of the composite, as predicted by analytical expressions and that obtained from numerical simulations, when the composite is under uniaxial plane strain loading ($\bar{\lambda}_1 = 1$, $\bar{\lambda}_2 = \bar{\lambda}$ and $\bar{\lambda}_3 = 1$). Stored energy density of the composite has been plotted as a function of the percentage applied macroscopic strain $\bar{\epsilon} = \bar{\lambda} - 1$. The results are plotted for initial porosity $f_o = 0.4, 0.6$ and power-law index $n = 0.4, 0.7, 1.3$ and 1.6 . Circular, square markers denote numerical results whereas solid, dashed lines denote analytical results. It is observed that the analytical results are an overestimate as compared to numerical results. Stored energy density remains unaffected by the power-law index of the fluid and is influenced only by the initial porosity in the elastomer. For a given porosity, the stored energy density curves for different values of n obtained from numerical simulations overlap. Therefore, the stored energy density in the composite remains unaffected by the nature of fluid.

5.3. Validation of dissipation density of the composite

Fig. 7 shows the variation of dissipation density of the density as estimated using analytical expressions and that obtained from numerical simulations. Macroscopic dissipation density is plotted as a function of the percentage strain $\bar{\epsilon}$ at loading rate $\dot{\lambda} = -0.05 \text{ s}^{-1}$ for initial porosities $f_o = 0.4, 0.6$ and power-law indices $n = 0.4, 0.7, 1.3, 1.6$. Here, $n = 0.4, 0.7$ imply shear-thinning fluids whereas, $n = 1.3, 1.6$ imply shear-thickening fluid. It is observed that the results obtained from numerical simulation match well with the analytical estimates for all the values of f_o and n . It can be seen that viscous dissipation density increases with the applied strain. The change in viscous dissipation with strain is higher for shear-thickening fluid as compared to that for shear-thinning fluid. Another important observation from Fig. 7(a) and Fig. 7(b) is that viscous dissipation density decreases with increase in porosity.

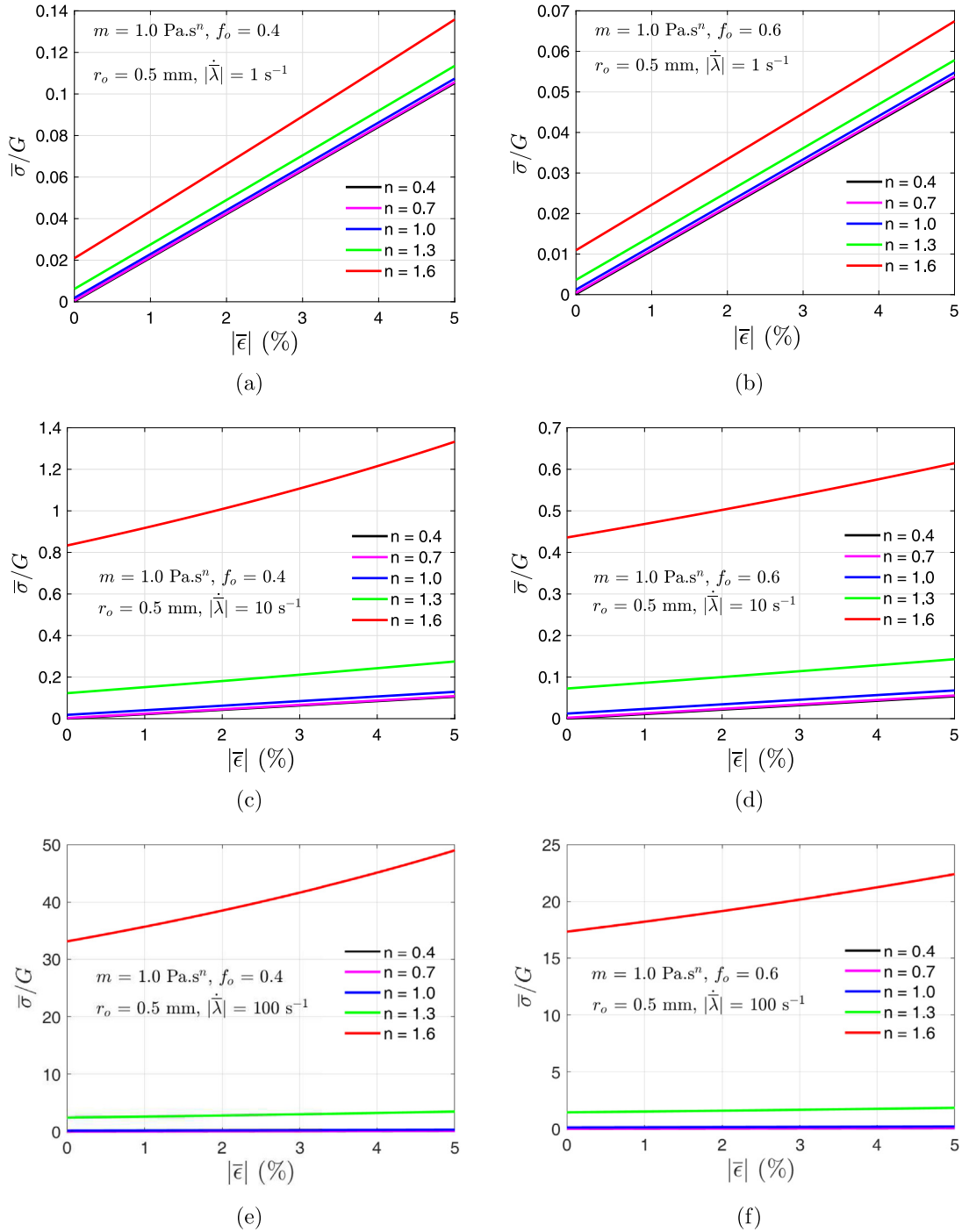


Fig. 8. Macroscopic response of power-law fluid filled elastomer subjected to monotonic compressive loading for $n = 0.4, 0.7, 1.0, 1.3$ and 1.6 , $f_o = 0.4, 0.6$ and applied load frequency $\omega = 1, 10$ and 100 rad/s.

5.4. Macroscopic response of the elastomeric composite

5.4.1. Monotonic loading

Fig. 8 shows macroscopic response of power-law fluid filled elastomer subjected to monotonic uniaxial plane strain compression. The results have been plotted for fluid consistency index $m = 1 \text{ Pa.s}^n$, initial pore radius $r_o = 0.5 \text{ mm}$, initial porosity $f_o = 0.4, 0.6$ and loading rate $|\dot{\lambda}| = 1 \text{ s}^{-1}, 10 \text{ s}^{-1}$ and 100 s^{-1} . Shear modulus of the elastomer has been used to normalize the macroscopic response so as to exclude the effect of elastomer material property on the macroscopic response of the composite. It is observed that loading rate strongly affects the

response of shear-thickening fluid filled elastomer in comparison to shear-thinning fluid filled elastomer. Further, the applied stress reduces with increases in initial porosity. It can be seen that for both $f_o = 0.4$ and 0.6 , as the loading rate increases, the response for $n = 1.6$ is much stiffer (≈ 10 times) than that for $n = 1.3$.

5.4.2. Harmonic loading

Fig. 9 shows the applied harmonic strain as a function of time and stress response of the composite as a function of applied strain. Stress-strain curves have been plotted for $f_o = 0.4$ and 0.6 to compare the effect of initial porosity on the overall response of the composite. The

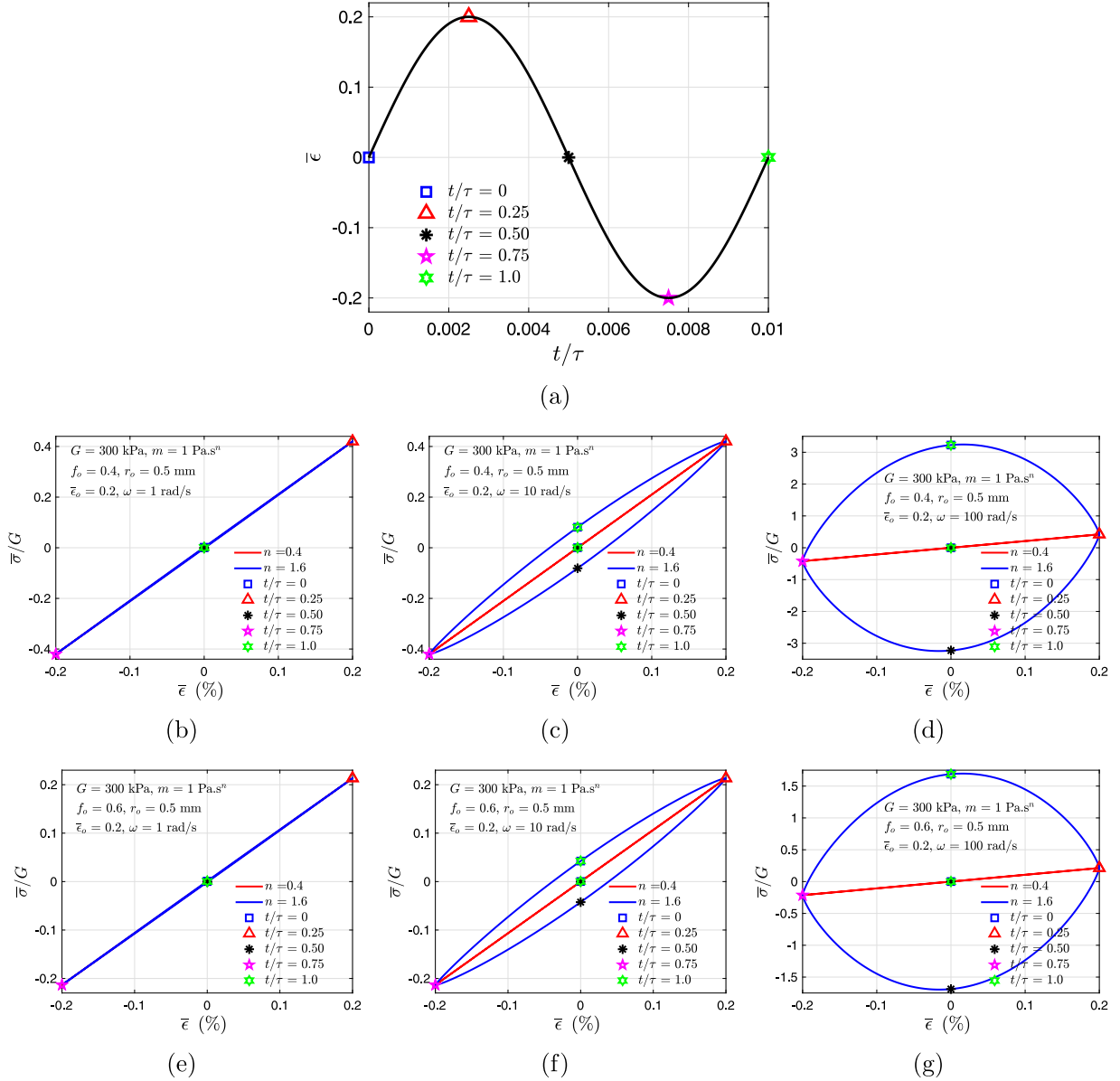


Fig. 9. (a) Applied harmonic strain as a function of time, (b, c, d) macroscopic response of the composite for $f_o = 0.4$, (e, f, g) macroscopic response of the composite for $f_o = 0.6$. Stress response has been plotted for $n = 0.4, 1.6$, $f_o = 0.6$ and 0.4 , $m = 1 \text{ Pa s}^n$, $\epsilon_o = 0.2a_o$ and $\omega = 1, 10$ and 100 rad/s . Markers show different time instances during the load cycle.

response values at $t/\tau = 0, 0.50$ and 1.00 can be attributed to the viscous term in (34), whereas the response at $t/\tau = 0.25$ and 0.75 is purely elastic in nature. $\tau = 2\pi/\omega$ is the time period of harmonic input. For $\omega = 1 \text{ rad/s}$ the responses for $n = 0.4, 1.6$ overlap as the response is dominated by the elastomer behavior. For $\omega = 10 \text{ rad/s}$ dissipation effect becomes noticeable as the response curve for $n = 1.6$ has a slight offset from the curve for $n = 0.4$ at $t/\tau = 0$ from dissipation in fluid. In the case of $\omega = 100 \text{ rad/s}$ a distorted sinusoidal with amplified peak magnitude is observed as the viscous dissipation starts to become dominant.

Conclusions

In the present study we have extended the framework earlier developed for estimating macroscopic behavior and Kelvin–Voigt parameters

for Newtonian fluid filled elastomers to a more general case of power-law fluid filled elastomer. The entire framework is based on micro-mechanical model and variational principle. Using the framework we derive semi-analytical estimates for average strain energy density in elastomer, dissipation in power-law fluid and the macroscopic behavior of the composite. Finite element analysis software COMSOL multi-physics was used to do the numerical simulations for validating the analytical estimates. The obtained estimates had an excellent match with the results from numerical simulation. The effect of parameters like initial porosity in the elastomer, power-law index of the fluid and loading rates on the macroscopic behavior of composite has been investigated. For a special case of uniaxial plane strain loading wherein the composite can be assumed to behave like according to Kelvin–Voigt model, expressions for its stiffness and damping coefficient have been

derived. The damping coefficient is a function of fluid consistency index, power-law index, pore half-length, initial pore radius and loading rate. Presented results can be used for design of custom composite materials for suspension, impact protection and vibration isolation applications.

CRediT authorship contribution statement

Vivek Singh: Conceptualization, Methodology, Software, Validation, Formal analysis, Visualization, Investigation, Writing – original draft, Writing – review & editing. **Vikranth Racherla:** Conceptualization, Methodology, Validation, Formal analysis, Visualization, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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